Q1. Given vertices **V = {A, B, C, D, E}** and edges
E = {(A,B), (A,C), (B,D), (C,E), (D,E)},
(a) Draw the graph.
(b) Write its **Adjacency Matrix**.
(c) Write its **Adjacency List** representation.

Q2. Consider the following **Adjacency Matrix** for a directed graph:

	A	B	C	D
A	0	1	1	0
B	0	0	1	1
C	0	0	0	1
D	0	0	0	0

(a) Write the **Edge List**.
(b) Find the **out-degree** and **in-degree** of each vertex.
(c) Comment on whether the graph is **acyclic** or not.

Q3. A graph has **6 vertices** and the following **Adjacency List**:

1 → 2, 3

2 → 1, 4, 5

3 → 1, 6

4 → 2

5 → 2, 6

6 → 3, 5

(a) Write the **set of edges**.
(b) Construct the **Adjacency Matrix** for this graph.
(c) Identify whether it is a **directed** or **undirected** graph.

Q4. The following **Edge List** is given for a graph:
E = {(1,2), (1,3), (2,3), (3,4), (4,5)}
(a) Draw the graph.
(b) Construct the **Adjacency Matrix**.
(c) How many vertices have **degree = 2**?
(d) Is the graph **connected**?

Q5. Given an **Adjacency Matrix** of size 4×4 as below:

	1	2	3	4
1	0	1	0	0
2	1	0	1	1
3	0	1	0	1
4	0	1	1	0

(a) List all **edges** in the graph.
(b) Write the **Adjacency List**.
(c) Calculate the **degree** of each vertex.
(d) Is the graph **sparse or dense**? Justify.

Q6. A graph has vertices **A, B, C, D, E** and its **Adjacency List** is:

A → B, D

B → A, C

C → B, E

D → A, E

E → C, D

(a) Draw the graph.
(b) List all **edges**.
(c) Construct its **Adjacency Matrix**.
(d) Determine whether the graph contains a **cycle**.